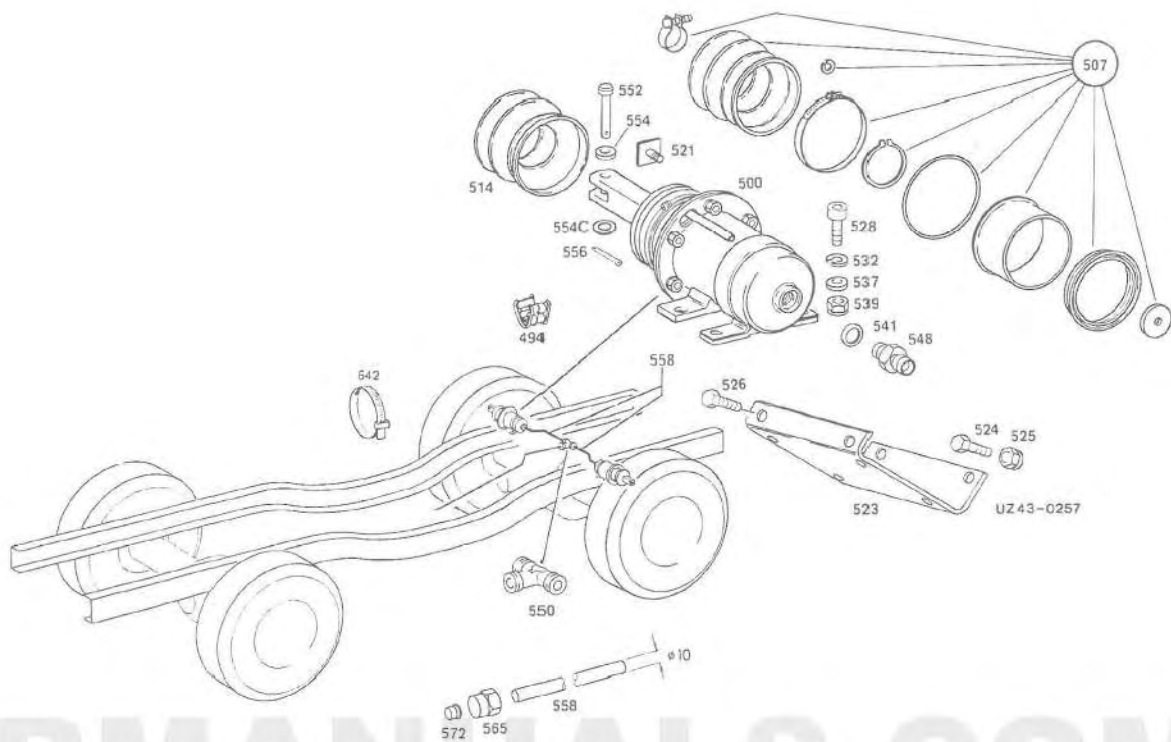


Exploded View



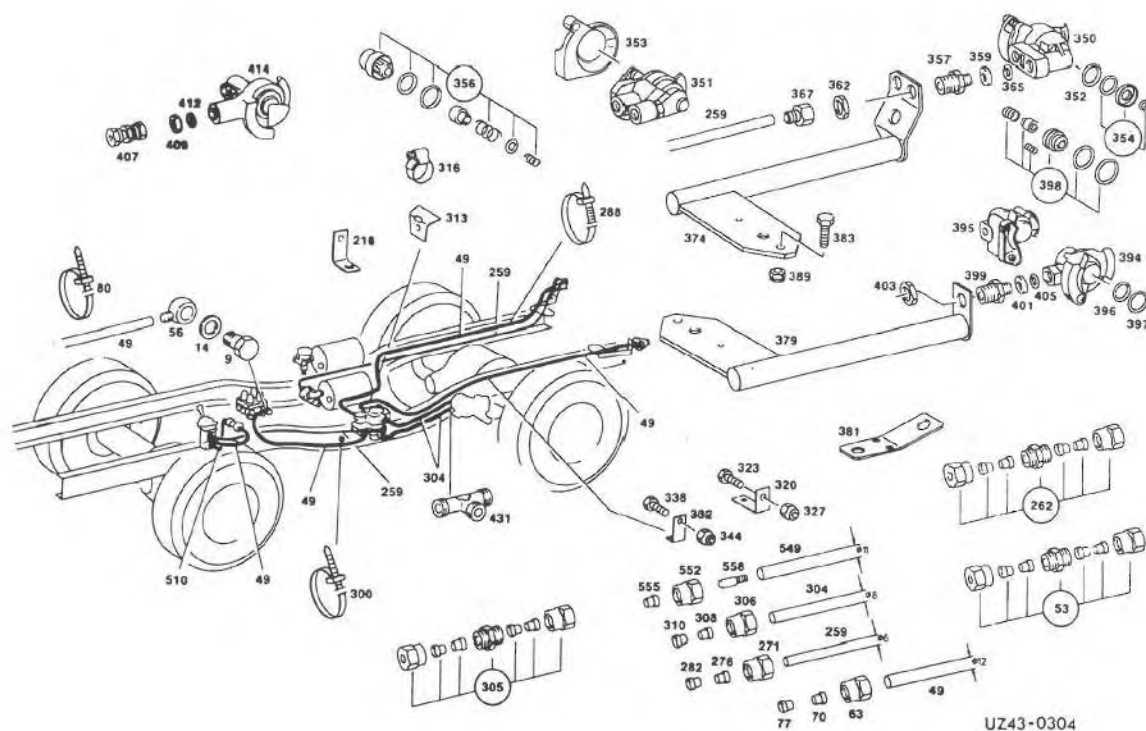
Spring-Loaded Parking Brake

494	Mount	539	Nut
500	Spring-loaded cylinder	541	Sealing ring
507	Repair kit	548	Pipe socket
514	Sleeve	550	Pipe socket
521	Bolt	552	Pin
523	Bracket	554	Washer
524	Bolt	554C	Washer
525	Nut	556	Cotter pin
526	Bolt	558	Line
528	Bolt	565	Union nut
532	Lock washer	572	Bite ring
537	Washer	642	Clip

43.11 General

SA 36 043

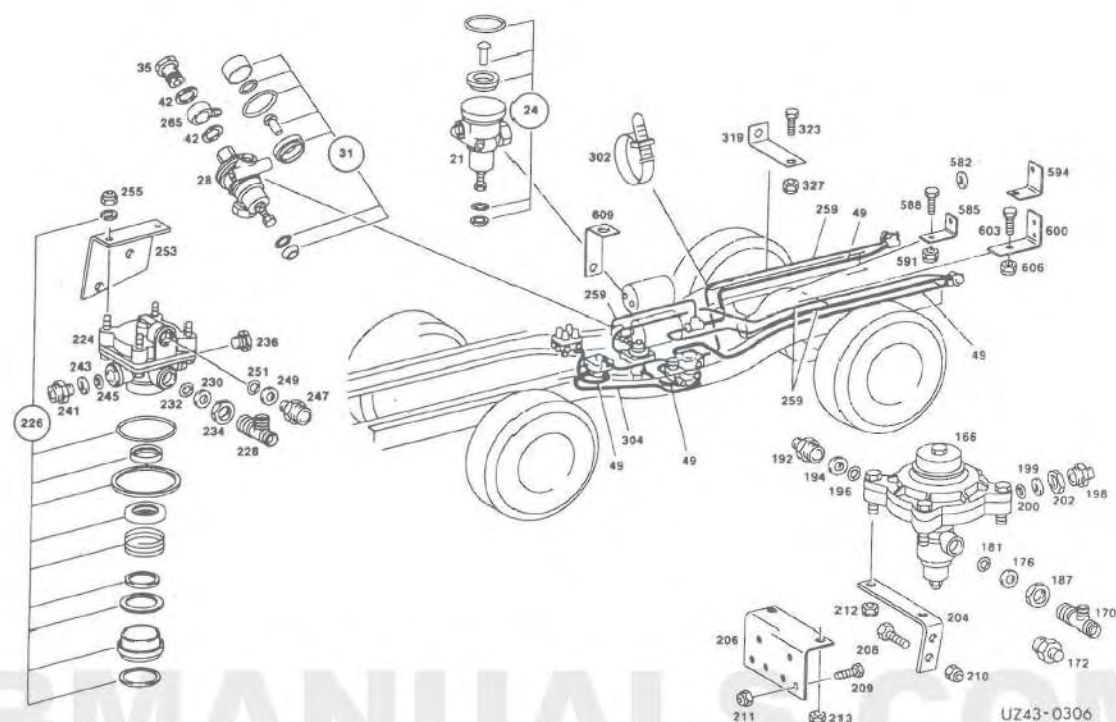
Exploded View



Trailer Brake System

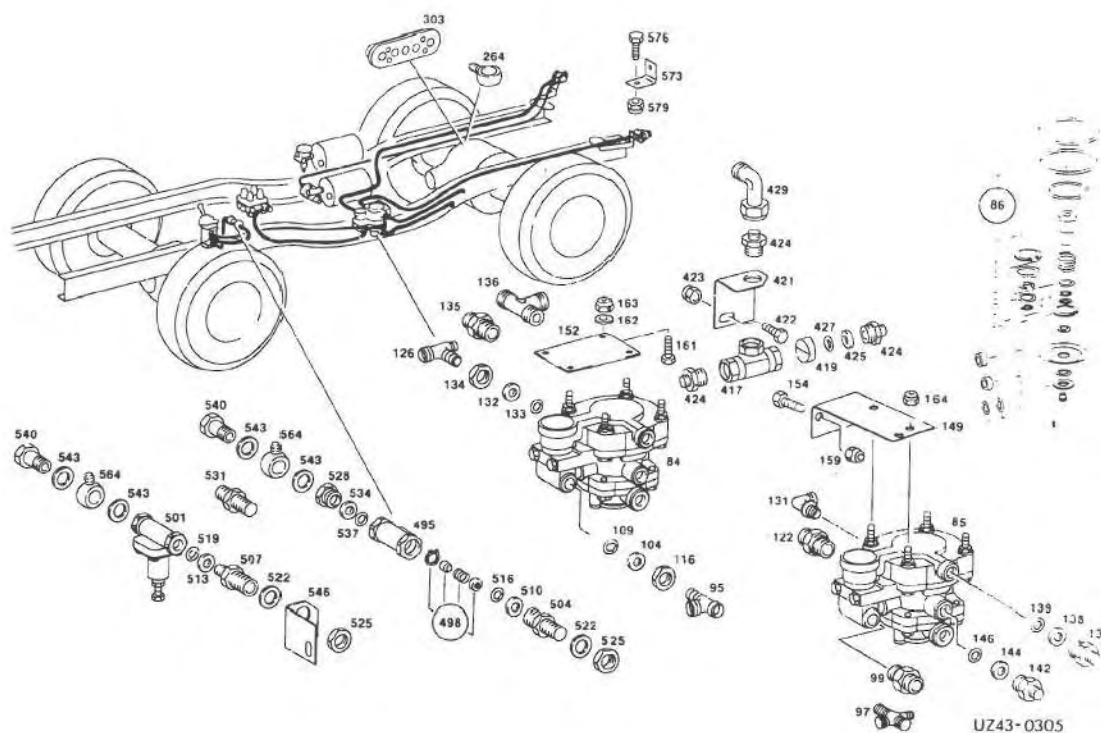
53	Repair kit for line breakage	379	Bracket
262	Repair kit for line breakage	381	Bracket
305	Repair kit for line breakage	394	Coupling head brake
350	Coupling head supply	395	Coupling head brake
351	Coupling head supply	396	Sealing ring
352	Sealing ring	397	Sealing ring
353	Cover	398	Repair kit for coupling head
354	Repair kit for coupling head	407	Pipe socket
356	Repair kit for coupling head	414	Coupling head single-line
357	Pipe socket	431	Pipe socket
374	Bracket		

Exploded View



Trailer Brake System

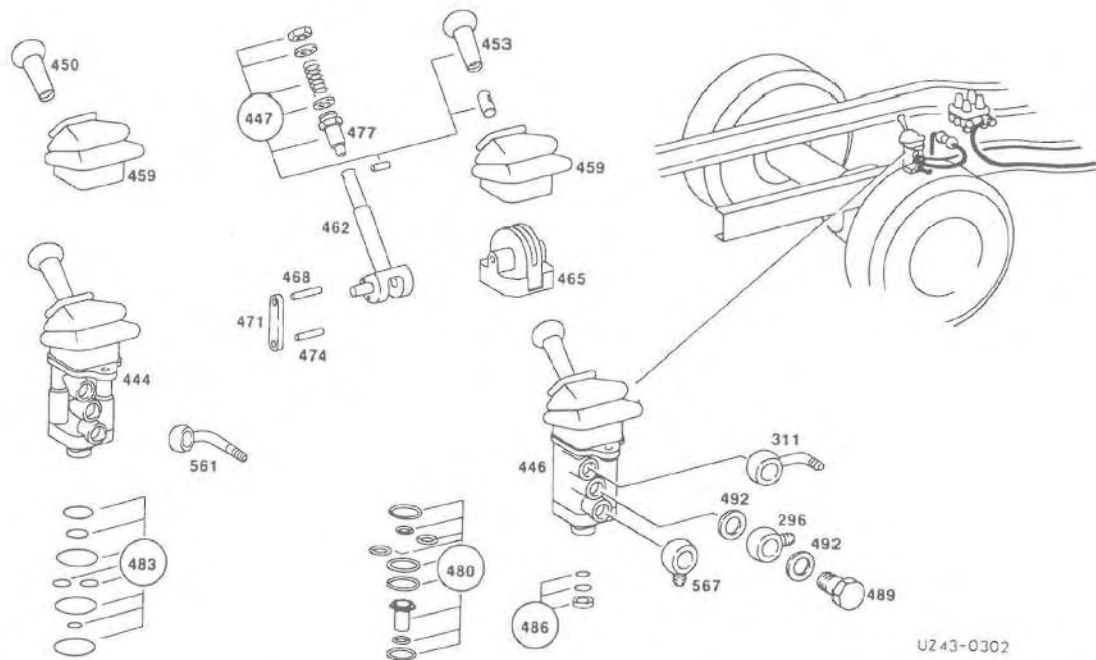
- 21 Pressure limiting valve
- 24 Repair kit for pressure limiting valve
- 28 Pressure limiting valve
- 31 Repair kit for pressure limiting valve
- 166 Trailer control valve, single-line
- 224 Relay valve
- 226 Repair kit for relay valve



Trailer Brake System

- 84 Trailer control valve, dual-line
- 85 Trailer control valve, dual-line
- 86 Repair kit for trailer control valve
- 417 Two-way valve
- 495 Non-return valve
- 498 Repair kit for non-return valve
- 501 Non-return valve with limited return flow

Exploded View



U2 43-0302

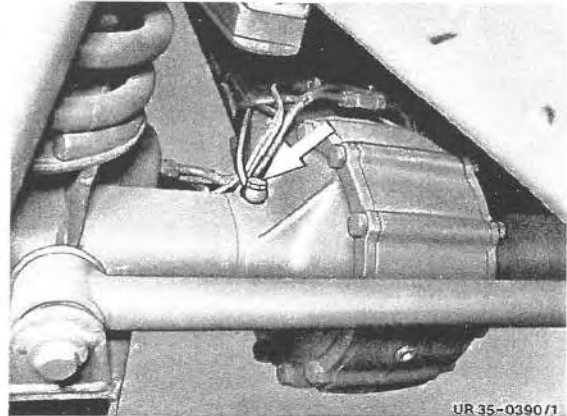
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Trailer Brake System

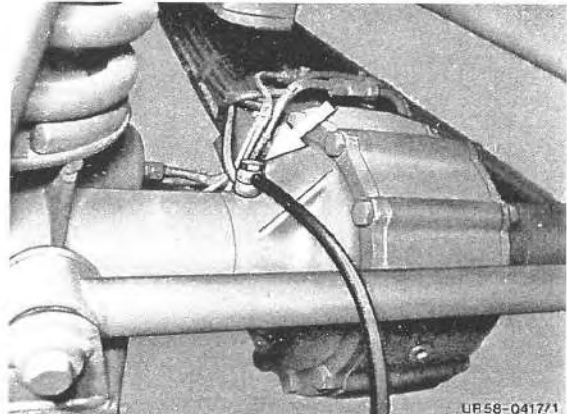
- 444 Parking brake valve
- 446 Parking brake valve
- 447 Grip
- 459 Sleeve
- 465 Cover
- 480 Repair kit for parking brake valve
- 483 Repair kit for parking brake valve

Checking and Adjusting

- 1 Secure vehicle against rolling away.
- 2 Release parking brake.
- 3 Disconnect air admittance and escape line at rear axle tube.



- 4 Connect test line consisting of a low pressure gauge (up to 1 bar) a plastic hose and an inlet union with banjo bolt to rear axle tube.



- 5 Apply supply pressure over 12 bar.

- 6 Activate four-wheel drive.

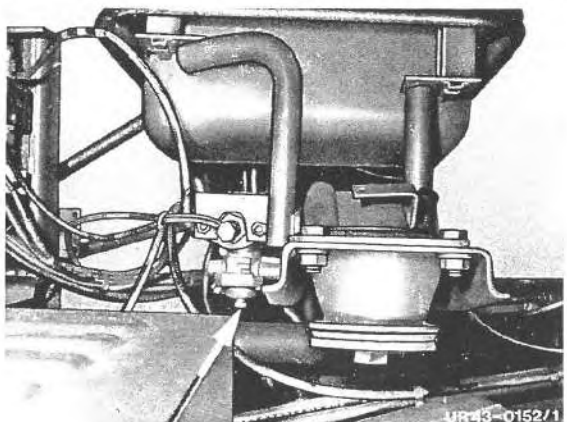


- 7 Check set operating pressure (0,2 bar) for truck operation or adjust pressure limiter if necessary.

Note:

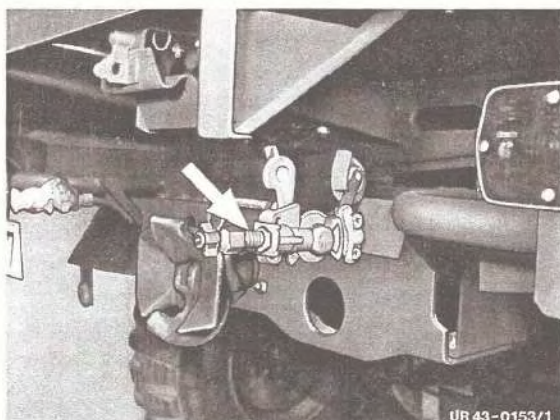
Pressure is increased by screwing in adjustment screw.
Pressure is decreased by screwing out adjustment screw.

- 8 Deactivate four-wheel drive.
Pressure must drop to 0 bar.

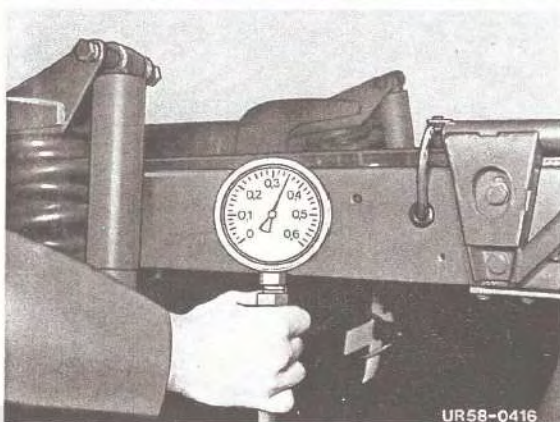


43.11 Checking and Adjusting Unit Operating Pressure

SA 35 976



9 Attach dummy coupling head to supply coupling head.



10 Once again switch on four-wheel drive.

11 Check set operating pressure (0,35 bar) for trailer drive using pressure gauge.

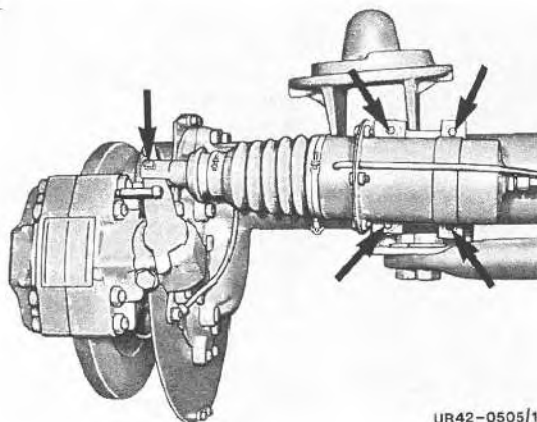
12 Repeat set value test for truck and trailer operation several times.

13 Remove tester and reconnect air admission and escape lines.

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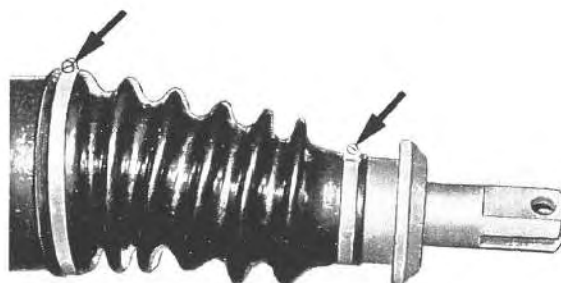
Renewing

- 1 Remove wheel.
- 2 Release spring-loaded cylinder and remove.



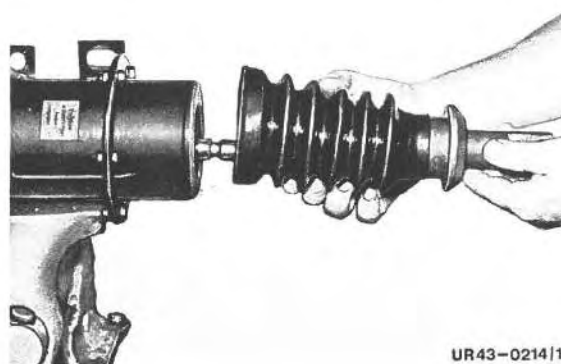
UR42-0505/1

- 3 Detach hose clips.



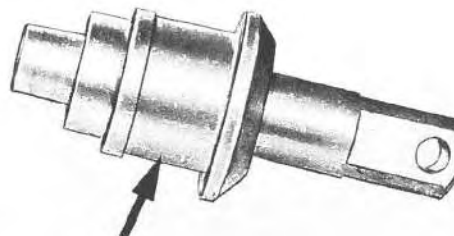
UR43-0213/1

- 4 Remove emergency release mechanism and bellows.



UR43-0214/1

- 5 Clean contact surface for bellows on emergency release mechanism.

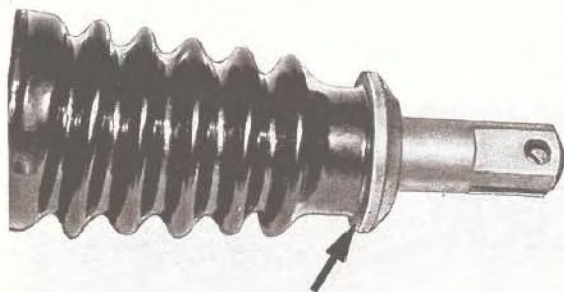


UR43-0215/1

43.11 Renewing Bellows on Spring-Loaded Cylinder

SA 35 966

- 6 Slide on new bellows up to the collar on the emergency release mechanism.

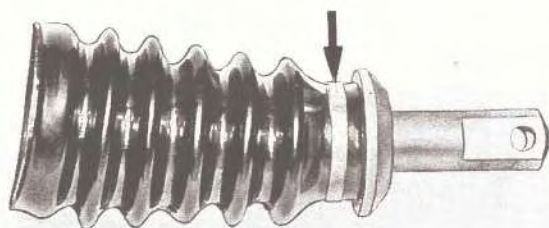


UR43-0216|1

- 7 Secure bellows on emergency release mechanism.

Note: New **clips** should be used for WABCO spring-loaded cylinders.

The old **hose clips with worm screw** can be used for GRAU spring-loaded cylinders.



UR43-0217|1

- 8 Coat locking balls with water pump grease and insert in emergency release mechanism.



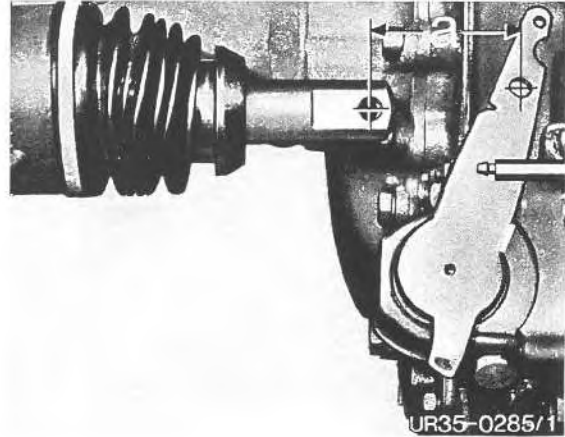
UR43-0218|1

- 9 Slide emergency release mechanism until it engages on the piston rod of the spring-loaded cylinder and secure bellows with hose clip at cylinder.

Note: Use new **clips** for WABCO spring-loaded cylinder.

The old **hose clips with worm screw** can be used for GRAU spring-loaded cylinders.

- 10 Install spring-loaded cylinder.
- 11 Adjust spring-loaded cylinder relevant to fixed caliper.
See 35.6 – 3.1/5.
- 12 Pressurize spring-loaded cylinder and attach to actuating lever.
- 13 Fit wheel.



Contents

Chapter	Page
1 General	
Installation Survey	1.1/1
Plastic Pipelines	1.2/1
Troubleshooting Compressed Air System	1.3/1
Brake Diagrams	1.4/1
Functional Diagrams	1.5/1
Functional Diagram, Auxiliary Load	1.6/1
Exploded Views	1.7/1
Renewing Gaiter of Spring Brake	2.1/1

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Installation Survey

Chassis		Brake System	
Model	Sales designation	Trailer	Installation
435.110	U 1700	Dual-line	SA 36 043
435.111	U 1700 L		
435.113	U 1700 L/38	Single and dual-line	
435.160	TM 170 Fa. Thyssen	—	

Notes on Plastic Pipelines

Special tools

Cons. No.	Tool	Special tool	Tool set
1	Clamp	387 589 02 37 00	B

Material:

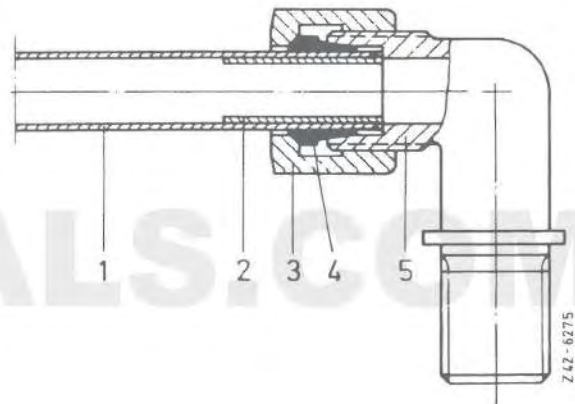
Plastic pipelines are made of polyamide 11 or 12 (W = soft, LT = light and thermally stable).

Polyamide is flexible and resists all fluids, oils and greases used in motor vehicles. In addition, plastic piping is also resistant to alkalines, unchlorinated solvents, organic and inorganic acids and diluted oxidents. The melting point of plastic piping is +186° C. Normal operating temperatures lie between -40° C and +100° C.

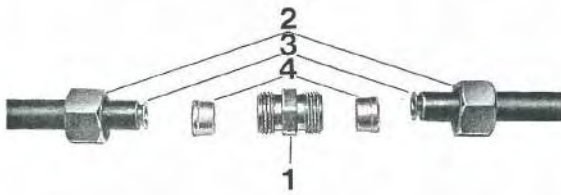
Plastic piping should be protected or removed prior to drilling and welding work.

Coupling and Fitting:

When preparing every new screw joint, the plastic pipe (1) must be cut at right angles. The sleeve (2) is then inserted as far as it will go in the end of the pipe, the union nut (3) and the bite ring (4) are then slid on to the plastic pipe. Now hand-tighten union nut until appreciable resistance is felt at pipe socket (5). Using wrench, tighten by a further 1 1/2 to 1 3/4 turns. Release union nut and check whether bite ring is seated firmly in plastic pipe and also whether a slight bulge has appeared in front of the bite ring. Retighten union nut in the normal manner.



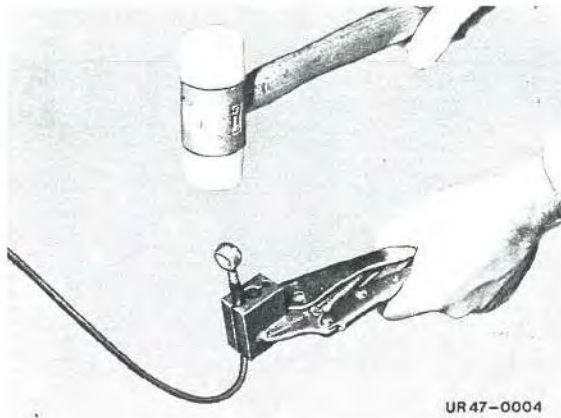
Damage to a plastic pipeline does not imply replacement of the entire line. Damaged sections can be cut out and repaired with the aid of a double adapter.



R47-6001

- 1 Double adapter
- 2 Nut
- 3 Sleeve insert
- 4 Bite ring

Care must be taken when disassembling and assembling **inlet union impact connections** to ensure that the sealing edges of the sawtooth-shaped retaining ribs are not damaged otherwise leaks may occur. Damaged connections must be renewed.



UR 47-0004

- 1 Hold plastic pipe in clamp.

- 2 Place inlet union connection on end of pipe and drive in as far as it will go with plastic-headed hammer.

Note:

The end of the pipe **must not** be heated up (room temperature).

The tip of the inlet union should be lightly oiled in the case of thin-walled or pipes of a small diameter thereby facilitating the entry of the inlet union.

Laying:

Plastic pipes must be arranged in such a way to ensure chafing cannot occur and secured with cable clips at 50 cm intervals. It must be possible to still shift the plastic pipe in the tightened cable clip.

When bending the pipe, a minimum radius must be observed otherwise the pipe may kink.

Minimum Bending Diameter for Plastic Pipes

Pipe size Outer diameter × wall thickness	Minimum permissible bending radius (r) in mm
4 × 1	20
6 × 1	30
6 × 0,5	40
8 × 1	40
8 × 2	40
12 × 1,5	60
13 × 1,5	65

Troubleshooting

Compressed air system (EEC system)

Repairs to assemblies may only be executed by authorized workshops and brake service points.

Fault	Cause	Remedy
Air compressor/auxiliary air compressor		
Compressed air delivery inadequate or non-existent.	Suction or delivery valves leaking. Excessive play between piston and cylinder. Line or filter in pressure regulator coked up. Cylinder head gasket defective.	Overhaul or exchange air compressor. Clean line or filter in pressure regulator, exchanging if necessary. Exchange cylinder head gasket.
Air compressor overheating.	Brake system leaking heavily.	Rectify leakage in brake system.
	Line or filter from pressure regulator coked up.	Clean line or filter from pressure regulator, exchanging if necessary.
	Too many additional air loads.	Install auxiliary air compressor.
Air compressor using too much oil	Vacuum in intake line.	Clean air filter, exchanging filter element if necessary.
	Cylinder and piston rings heavily worn.	Overhaul or exchange air compressor.
Pressure regulator		
At charging position, compressed air escapes at outlet socket.	No load valve leaking.	Clean or exchange valve and valve seat.
Pressure regulator fails to switch air compressor over to no-load.	Diaphragm piston sticking.	Check or exchange diaphragm piston.
Pressure regulator fails to switch air compressor over to on-load.	Adjusting spring defective, cut-off piston sticking.	Exchange spring, check cut-off piston and sealing ring.
At charging position, pressure regulator loses compressed air at vent.	Intake valve leaking, spring defective.	Clean or exchange valve, exchange spring.
At no-load position, pressure regulator loses compressed air at vent.	Exhaust valve leaking, spring defective.	Clean or exchange valve, exchange spring.
Pressure regulator does not admit compressed air to air tank.	Filter choked.	Clean or exchange filter.
Brief switching interval between pressure regulator cut-in and cut-out.	High air consumption, check valve leaking, sealing ring or exhaust valve leaking.	Check brake system for leakage. Check valves and seals. Exchange wear parts.

Fault	Cause	Remedy
Defroster		
Compressed air escapes at actuating rod.	Sealing ring leaking.	Exchange sealing ring at actuating rod.
At closed position, steady consumption of alcohol.	Sealing ring leaking.	Exchange sealing ring.
Defroster does not work satisfactorily at operating position.	No antifreeze left. More water than antifreeze in supply tank.	Drain water and fill up with antifreeze.
Four-circuit protection valve		
Compressed air escapes at breather hole.	Seal in valve body leaking.	Check or replace seal.
Opening pressure of individual valve circuits not within tolerance range.	Incorrect adjustment	Check valve setting on test bench. Replace spring and set on test bench.
	Spring defective.	
With failure in first or second circuit, pressure not maintained in third or fourth circuit.	Valve body sticking or jammed.	Exchange 4-circuit protection valve.
	Check valves leaking or defective.	Check or exchange check valves.
Drain valve		
Valve leaking.	Actuating valve seat dirty or defective.	Briefly operate actuating valve. Exchange if valve still leaks.
Check valve with limited return flow		
Valve does not hold back closing pressure.	Diaphragm or check valve leaking.	Check diaphragm or valve, replacing if necessary.
Double pressure gauge		
Both pointers do not agree when cut-off pressure reached, out of tolerance ± 0.2 bar.	Wear in transmission mechanism.	Exchange double pressure gauge.
Air escapes at double pressure gauge.	Solder point leaking.	Exchange double pressure gauge.
Warning light		
Light comes on while travelling.	Tank pressure inadequate or non-existent.	Watch double brake pressure gauge until tank pressure is reached. Check compressed air system for leaks, sealing if necessary.
CAUTION: Stop immediately		
	Leakage in hydraulic brake circuit.	Rectify leakage, bleeding brake system if necessary. Check brake reservoir, correcting level if necessary.

Fault	Cause	Remedy
	Brake pad/lining wear	Exchange brake pads/linings. Check brake reservoir, correcting level if necessary.
Warning light does not come on when pressure drops below warning level.	Bulb defective.	Exchange bulb.
	Fault in electrical system.	Rectify electrical fault.
Dual-circuit service brake valve		
In release position, compressed air escapes at vent.	Intake valve leaking.	Exchange intake valve.
In partial and full braking positions, compressed air escapes at vent.	Exhaust valve or sealing rings leaking.	Exchange exhaust valve or sealing rings, exchanging brake valve if necessary.
In lower partial braking position, air escapes from vent.	Sealing ring in actuating cylinder of piston circuit 1 leaking.	Exchange actuating cylinder.
After braking, wheel brakes are slow to release or residual pressure remains in brake system.	No pedal play.	Adjust pedal play.
	Inner stiffness between piston and cylinder bore as well as at fulcrum point of angle lever due to defective bellows, compression weak.	Check brake valve, exchanging if necessary.
Brake graduation not fine enough.	Pedal system, cable connection stiff.	Release pedal system and cable connection.
	Inner stiffness between cylinder and piston bore.	Check brake valve, exchanging if necessary.
Parking brake valve		
At release position, compressed air escapes at vent or bellows.	Exhaust valve or sealing rings leaking.	Exchange exhaust valve and sealing rings.
At partial and full braking positions, compressed air escapes at vent.	Intake valve or sealing ring leaking.	Clean or exchange intake valve. Exchange sealing ring.
Valve does not give satisfactory graduation.	Vent dirty.	Clean vent.
	Inner stiffness at piston.	Overhaul valve, exchanging if necessary.
	Valve stiff at fulcrum point of hand lever.	Release hand lever, exchanging valve if necessary.
	Bellows defective.	Exchange bellows.
In release position, handbrake valve does not provide full air admission.	Poor valve adjustment.	Adjust handbrake valve on test bench.

Fault	Cause	Remedy
Trailer control valve, single-line		
With brakes released, compressed air escapes at vent and at control port.	Exhaust valve leaking, sealing rings leaking.	Clean or exchange valve and valve seat. Exchange sealing ring.
At closed position, compressed air escapes at vent.	Exhaust valve leaking, diaphragm defective.	Clean or exchange valve and valve seat. Exchange diaphragm.
Valve does not provide satisfactory graduation.	Graduating piston sticking. Sealing ring leaking. Snap ring defective.	Check piston and sealing ring, exchanging if necessary. Exchange snap ring.
Assembly does not provide correct advance.	Graduating piston sticking. Spring broken. Diaphragm defective.	Check piston and sealing ring. Exchange spring. Exchange diaphragm.
In release position, pressure in trailer control line rises above 5.4 bar.	Pressure reducing piston or spring defective, intake valve may be leaking.	Check piston and sealing ring, exchanging if necessary. Exchange spring. Check valve and valve seat, exchanging if necessary.
In braking position, pressure rises in trailer control line.	Intake valve leaking.	Check valve and valve seat, exchanging if necessary.
Trailer control line, dual-line		
With service and parking brake released, compressed air escapes at vent.	Intake valve or sealing rings leaking.	Clean or exchange intake valve, exchange sealing rings.
With service and parking brake released, compressed air escapes at vent of service brake valve.	Diaphragm piston leaking to control connection 42.	Exchange control valve.
Response stage of valve too great or graduation not proper. Residual pressure remains in trailer brake line.	Inner stiffness in valve (pistons sticking).	Checking control valve, exchanging if necessary.
In full and partial braking position, compressed air escapes at vent.	Exhaust valve or sealing rings leaking.	Clean or exchange exhaust valve. Exchange sealing rings.
Trailer control valve, dual-line		
With service and parking brake released, residual pressure remains in trailer control line.	Service brake valve does not fully exhaust.	Check pedal play and adjust. Check service brake valve, exchanging if necessary.
	Cable connection stiff.	Release cable connection.
	Handbrake valve does not fully admit air.	Check handbrake valve, exchanging if necessary.
With parking brake applied, compressed air escapes from vent.	Exhaust valve or sealing ring leaking.	Clean or exchange exhaust valve. Exchange sealing rings.

Fault	Cause	Remedy
Piston cylinder (actuating cylinder)		
In braking position, compressed air escapes at vent.	Piston sealing rings leaking.	Exchange sealing rings.
	Cylinder liner scored.	Exchange cylinders.
In partial braking position, compressed air escapes at vent of service brake valve.	Sealing ring at piston circuit 1 of actuating cylinder is leaking.	Exchange sealing ring, replacing cylinder if necessary.
Following braking, piston does not return to released position.	Piston sticking, compression spring defective.	Exchange cylinder.
Spring loaded cylinder		
With parking brake released, compressed air escapes at vent of spring loaded cylinder.	Sealing ring leaking.	Exchange sealing ring or exchange spring loaded cylinder.
Stroke of spring loaded cylinder excessive.	Air gap between brake pad and disk excessive.	Adjust air gap.
Spring loaded cylinder does not return fully to released position.	Release pressure inadequate.	Check parking brake, exchanging if necessary.
Two-way valve		
Compressed air escaping at non-modulated pressure connection line.	Piston slide valve leaking.	Exchange piston valve.
Pressure reducing valve		
Compressed air escapes at vent.	Sealing ring leaking.	Renew sealing ring or exchange valve.
Closing pressure of valve excessive or inadequate.	Spring bias reduced or sealing ring compressed.	Set closing pressure at adjusting screw.
Valve no longer closes. Over-pressure valve opens.	Spring or sealing ring defective. Piston sticking.	Exchange pressure reducing valve.
Coupling head, single-line		
Compressed air escapes in coupled position.	Sealing surfaces defective.	Exchange sealing ring.
Compressed air escapes at coupling head in uncoupled position.	Valve leaking.	Check valves and valve seats or exchange.
Insufficient compressed air to trailer.	Pin of counter coupling head does not press valve fully open.	Renew coupling head with pin.
	Filter clogged.	Clean filter.

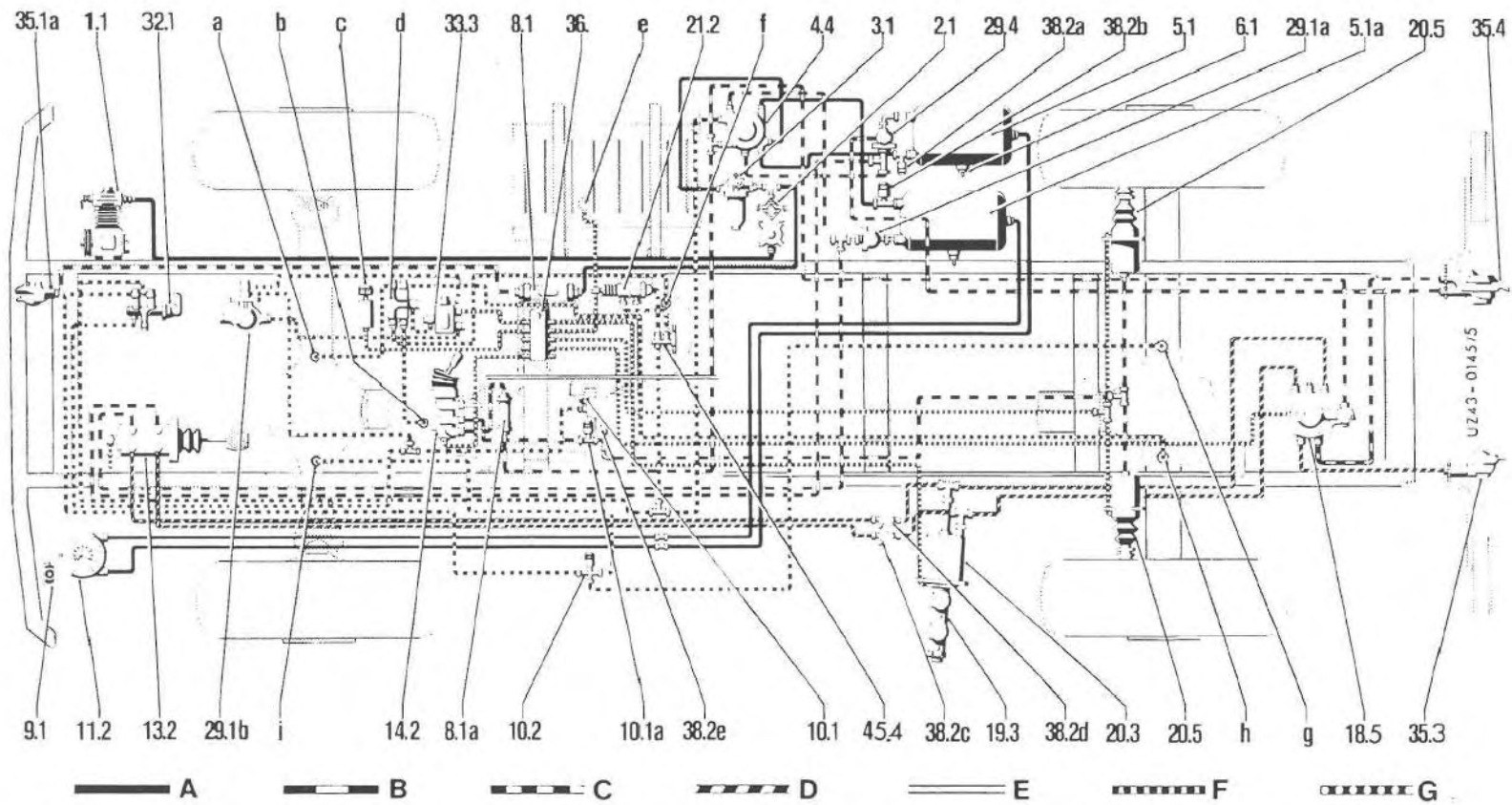
Fault	Cause	Remedy
Compressed air escapes at bore of valve of coupling head (special version) in uncoupled position.	Valve body leaking.	Clean valve or exchange coupling head.
Coupling heads, dual-line		
Following coupling, compressed air escapes at sealing surfaces during full braking action.	Sealing surfaces defective.	Exchange sealing rings.
Coupling heads do not lock properly.	Tab washer worn.	Exchange tab washers at both coupling heads.
Following uncoupling, air escapes at supply head.	Valve leaking.	Clean or exchange valve.
	Filter clogged.	Clean filter.
2/2-way valve (breadaway valve)		
Compressed air escapes at aspiration port.	Valve leaking.	Exchange valve.
Pressure drop in supply line during full braking in 2 seconds to 1.5 bar not achieved.	Restriction in valve too big or piston sticking.	Check or exchange valve.
3/2-way valve (actuating valve)		
In release position, compressed air escapes at vent.	Valve leaking.	Clean or exchange valve.
In actuating position, compressed air escapes continuously at vent.	Valve or sealing rings defective.	Clean valve, exchanging if necessary.
Actuating tappit does not fully return to released position.	Spring defective, tappit sticking.	Check or exchange valve.
3/2-way valve (test valve)		
With handbrake in released position, compressed air escapes at rotary knob.	Sealing ring on top of tappit leaking.	Exchange sealing ring or valve.
Handbrake exhausted. Compressed air escapes continuously at vent of handbrake valve.	Sealing ring on bottom of tappit leaking.	Exchange sealing ring or valve.
Handbrake exhausted. Air admitted to test valve. Compressed air escapes continuously at vent of handbrake valve.	Sealing ring in centre of tappit leaking.	Exchange sealing ring or valve.

Fault	Cause	Remedy
4/2-way valve (planetary gear)		
With slow or fast range selected, compressed air escapes at vent.	Sealing rings leaking.	Exchange sealing rings or valve.
	Piston cylinder on transmission leaking.	Exchange piston cylinder or sealing ring.
Actuating tappit does not return fully to released position.	Spring defective, tappit sticking.	Check or exchange valve.
Vehicle is parked in the evening with fast range selected. On the following morning the driver starts the vehicle, shifts into the slow range and drives off. If the ancillary consumers have not yet been supplied with air via the 4-circuit protection valve, the driver is travelling in the fast range although the 4/2-way valve has shifted to the slow range. As soon as the ancillary consumers are supplied with air, the planetary gear shifts four gears into the slow range which may result in consequential damage.	Pressure drop overnight in compressed air supply.	Do not move off until vehicle is ready for the road and ancillary consumers have supply pressure.
4/2-way valve (planetary gear)		
Vehicle is parked in the evening with the fast range selected. The next morning the driver starts the engine and selects the slow range. As a result, sufficient air was in the supply line to the 4/2-way valve to enable the fast range to be disengaged but not the slow range to be engaged. The driver selects a gear at the main transmission with the engine running, the supply reservoirs are filled and he leaves the cab. He has not noticed that the interruption of the power flow to the wheels was only at the planetary range which was not engaged. If the ancillary consumers now get supply pressure, the slow range is correctly selected and the vehicle moves off by itself.	Pressure drop overnight in compressed air supply.	After the engine is running and the compressed air system filled, do not select any gear in the main transmission.
CAUTION: Risk of accident !		

Fault	Cause	Remedy
4/3-way valve (relay valve)		
In released position, compressed air escapes at vent.	Sealing ring at bottom leaking.	Replace sealing ring or exchange valve.
With 4-wheel drive or differential locks engaged, compressed air escapes at vent.	Sealing ring in centre or top leaking.	Replace sealing ring or exchange valve.
Valves does not return to release position.	Spring broken, tappit sticking.	Check or replace valve.

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Brake Diagram (Special Vehicle 435.113)



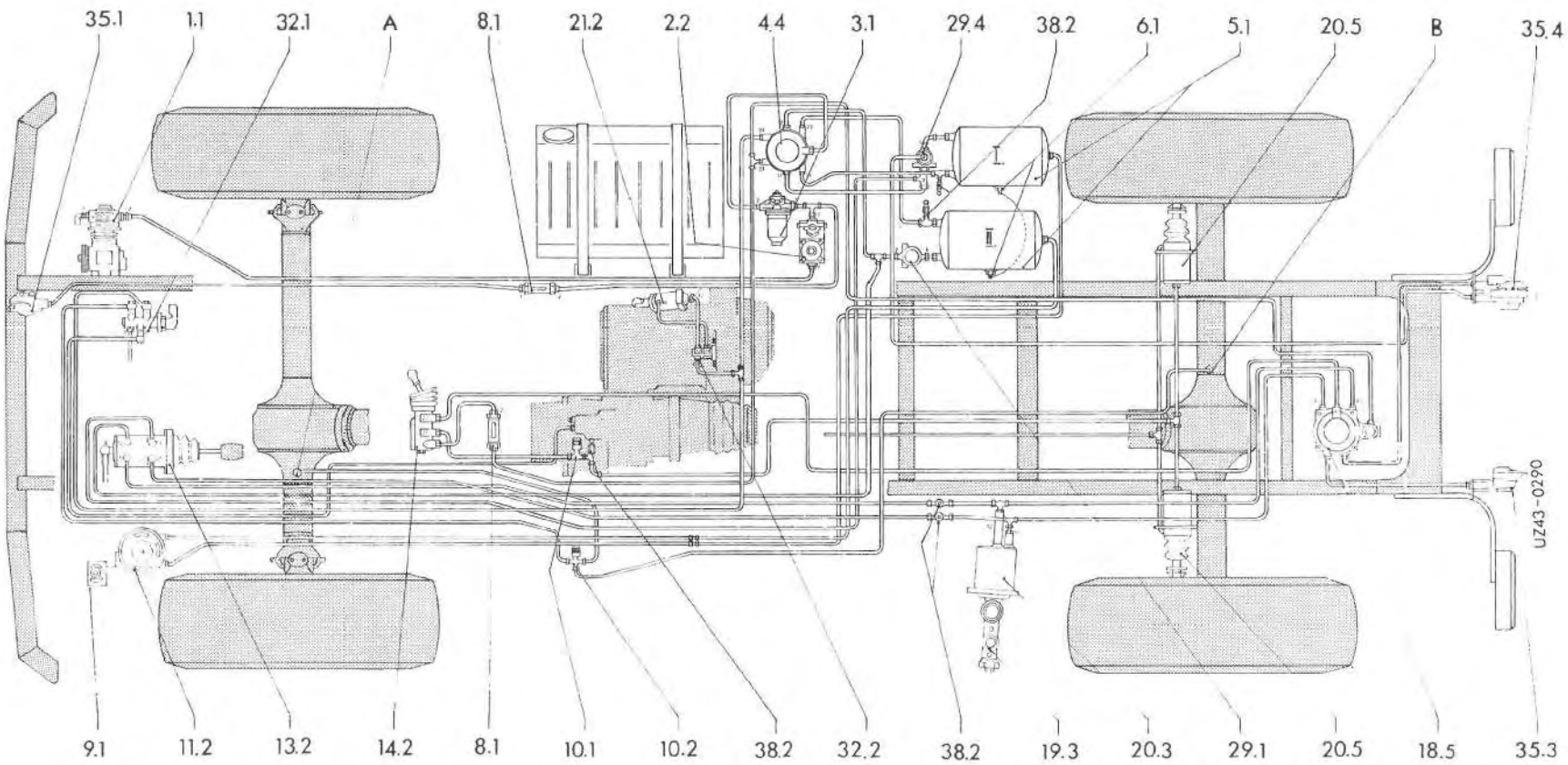
Dual-circuit air brake system, spring-loaded parking brake and dual-line trailer brake system.

Refer to 43.11 – 1.5/1 to 1.5/28 for symbols.

Key to 1.4/1

- | | | | |
|---|------------------------------------|---|---|
| A | High pressure supply 18 bar | a | Connection, front axle air admission and escape |
| B | Supply, brakes 9.5 and 7.0 bar | b | Connection, PTO transmission air admission and escape |
| C | Auxiliary load 9.5 bar and 7.0 bar | c | Starting pilot vent |
| D | Brake control pressure | d | Compressed air distributor |
| E | Trailer brake test pressure | e | Fuel tank vent |
| F | Air admission | f | Connection, transmission air admission and escape |
| G | Central vent | g | Connection, differential lock rear axle |
| | | h | Connection, rear axle air admission and escape |
| | | i | Connection, differential lock front axle |
-
- 1.1 Air compressor
 - 2.1 Pressure regulator with tyre inflating port
 - 3.1 Antifreeze unit
 - 4.4 Four-circuit protection valve
 - 5.1 Air tank (bottom)
 - 5.1a Air tank (top)
 - 6.1 Drain valve
 - 8.1 Non-return valve, charging line
 - 8.1a Non-return valve, spring-loaded cylinder
 - 9.1 Warning lamp, supply pressure
 - 10.1 Electrical pressure switch 4-wheel brake system
 - 10.1a Electrical pressure switch parking brake
 - 10.2 Electrical pressure switch differential lock
 - 11.2 Double pressure gauge (supply)
 - 13.2 Service brake valve (dual-circuit)
 - 14.2 Parking brake valve with brake test point
 - 18.5 Trailer control valve (2-line) with breakaway valve
 - 19.3 Master brake cylinder (dual-circuit)
 - 20.3 Preload cylinder (dual-circuit)
 - 20.5 Spring-loaded parking brake
 - 21.2 Shift cylinder planetary gear
 - 29.4 Pressure reducing valve can be switched over to trailer operation 9.5 bar, trailer operation 7.0 bar circuit 1
 - 29.1a Pressure reducing valve circuit 2
 - 29.1b Pressure reducing valve air admission
 - 33.3 3/2-way valve air admission
 - 35.1a Coupling head charging connection
 - 35.3 Coupling head brake
 - 35.4 Coupling head supply with shut-off valve
 - 36. Central vent filter
 - 38.2a Test connection supply pressure circuit 1
 - 38.2b Test connection supply pressure circuit 2
 - 38.2c Test connection brake pressure circuit 1
9.5 bar truck operation,
7.0 bar trailer operation
 - 38.2d Test connection brake pressure circuit 2
 - 38.2e Test connection spring-loaded cylinder
 - 42. Switchover valve 4-wheel differential lock
 - 42.a Switchover valve planetary gear

Brake Diagram (435.110/111/113)



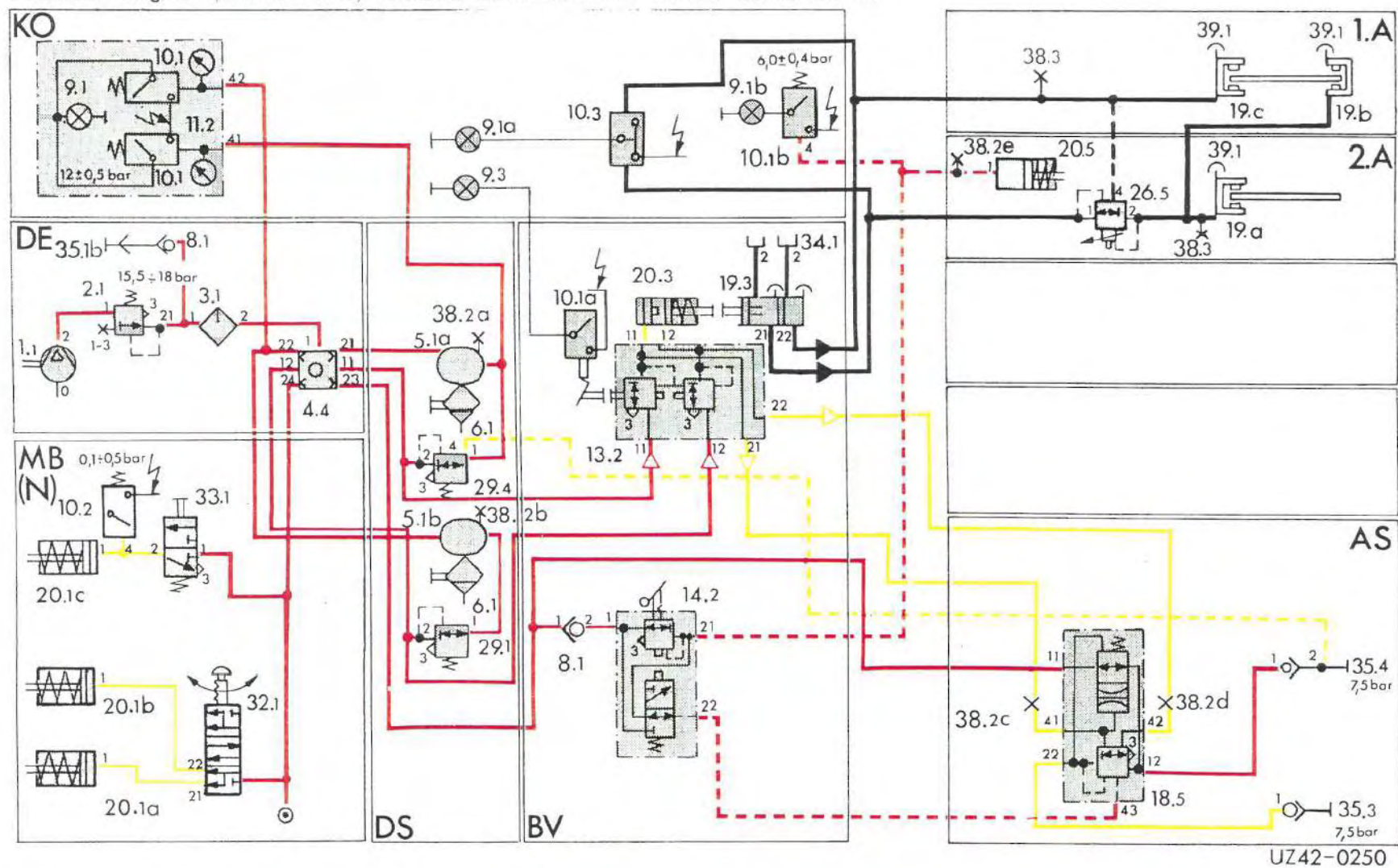
Dual-Circuit air system and dual-line trailer brake system, spring-loaded parking brake and auxiliary load.

Explanations to 1.4/3

1.1	Single cylinder air compressor	20.3	Precharging cylinder, dual-circuit
2.2	Pressure regulator	20.5	Spring-loaded brake cylinder
3.1	Defrosting pump	21.2	Shift cylinder
4.4	Four-circuit protective valve	29.1	Pressure reducing valve
5.1	Air tank	29.4	Pressure reducing valve, reversible
6.1	Drainage valve (water separator)	32.1	Relay valve, four-wheel drive differential locks
8.1	Check valve	32.2	4/2 -way valve
9.1	Warning light, red	35.1	Coupling head, supply
10.1	Pressure switch, parking brake	35.3	Coupling head, brake
10.2	Pressure switch, differential lock	35.4	Filler line for supply, front
11.2	Instrument cluster	38.2	Test connections, brake pressure, circuit I and II
13.2	Service brake valve	38.2	Test-and filler connections, air tank
14.2	Hand brake valve		
18.5	Trailer control valve with break-away valve	A	Connection differential lock, front
19.3	Master brake cylinder	B	Connection differential lock, rear

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Functional Diagram (435.110/111/113) To chassis end No. 435.110/111 086 854 435.113: 084 641



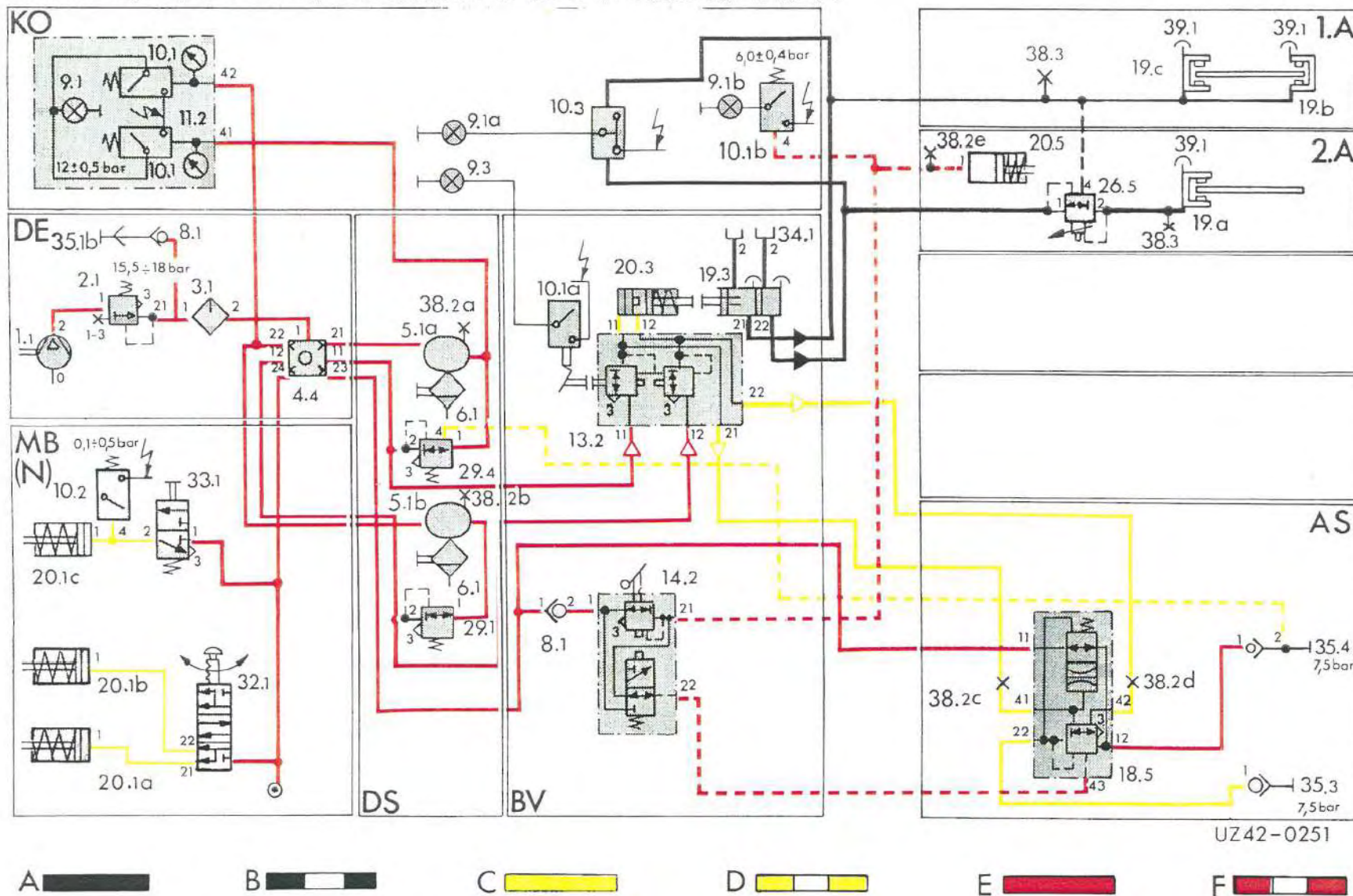
A B C D E F

Dual-circuit hydraulic system and dual-circuit air brake system, ALB on rear axle and $\frac{1}{2}$ front axle, spring-loaded parking brake and dual-line trailer brake system.

Key to 1.5/1

A	Hydraulics	1.A	= 1st. axle
B	Hydraulic control pressure	2.A	= 2nd axle
C	Brake pressure	AS	= Trailer control
D	Brake control pressure	BV	= Actuating valve
E	Supply pressure	DS	= Accumulator
F	Supply control pressure	MB(N)	= Engine brake (auxiliary load)
		DE	= Pressure generation
		KO	= Control switch
1.1	Air compressor	20.1 a	Piston cylinder 4-wheel drive
2.2	Pressure regulator with tyre inflating port	20.1 b	Piston cylinder differential lock
3.1	Antifreeze unit	20.1 c	Piston cylinder engine brake
4.4	Four-circuit protection valve	20.3	Preload cylinder
5.1a	Air tank circuit 1	20.5	Spring-loaded parking brake
5.1b	Air tank circuit 2	26.5	Automatic brake pressure modulator (ALB)
6.1	Drain valve	29.1	Pressure reducing valve
8.1	Non-return valve	29.4	Pressure reducing valve with pneumatic control
9.1	Warning lamp supply pressure	32.1	Switchover valve 4-wheel drive differential lock
9.1a	Indicator lamp pressure difference (SA 35 975)	33.1	Switchover valve engine brake
9.1b	Indicator lamp parking brake	34.1	Equalizing reservoir
9.3	Stoplight	35.1 b	Coupling head, front charging line
10.1	Electrical pressure switch supply pressure	35.3	Coupling head rear, brake
10.1a	Electrical pressure switch stoplight	35.4	Coupling head rear, supply
10.1b	Electrical pressure switch spring-loaded cylinder	38.2a	Test connection circuit 1
10.2	Electrical pressure switch engine brake	38.2b	Test connection circuit 2
10.3	Electrical pressure difference warning switch (SA 35 975)	38.2c	Test connection circuit 1
11.2	Double pressure gauge	38.2d	Test connection circuit 2
13.2	Service brake valve	38.2e	Test connection spring-loaded cylinder
14.2	Parking brake valve with brake test point	38.3	Test connection ALB modulator
18.5	Trailer control valve (2-line) with breakaway valve	39.1	Vent
19.a	Fixed combined caliper RA		
19.b	Fixed caliper FA circuit 1		
19.c	Fixed caliper FA circuit 2		
19.3	Master brake cylinder (dual-circuit)		

Functional Diagram (435.110/111/113) From chassis end No. 435.110/111: 086 055 435.113: 084 642



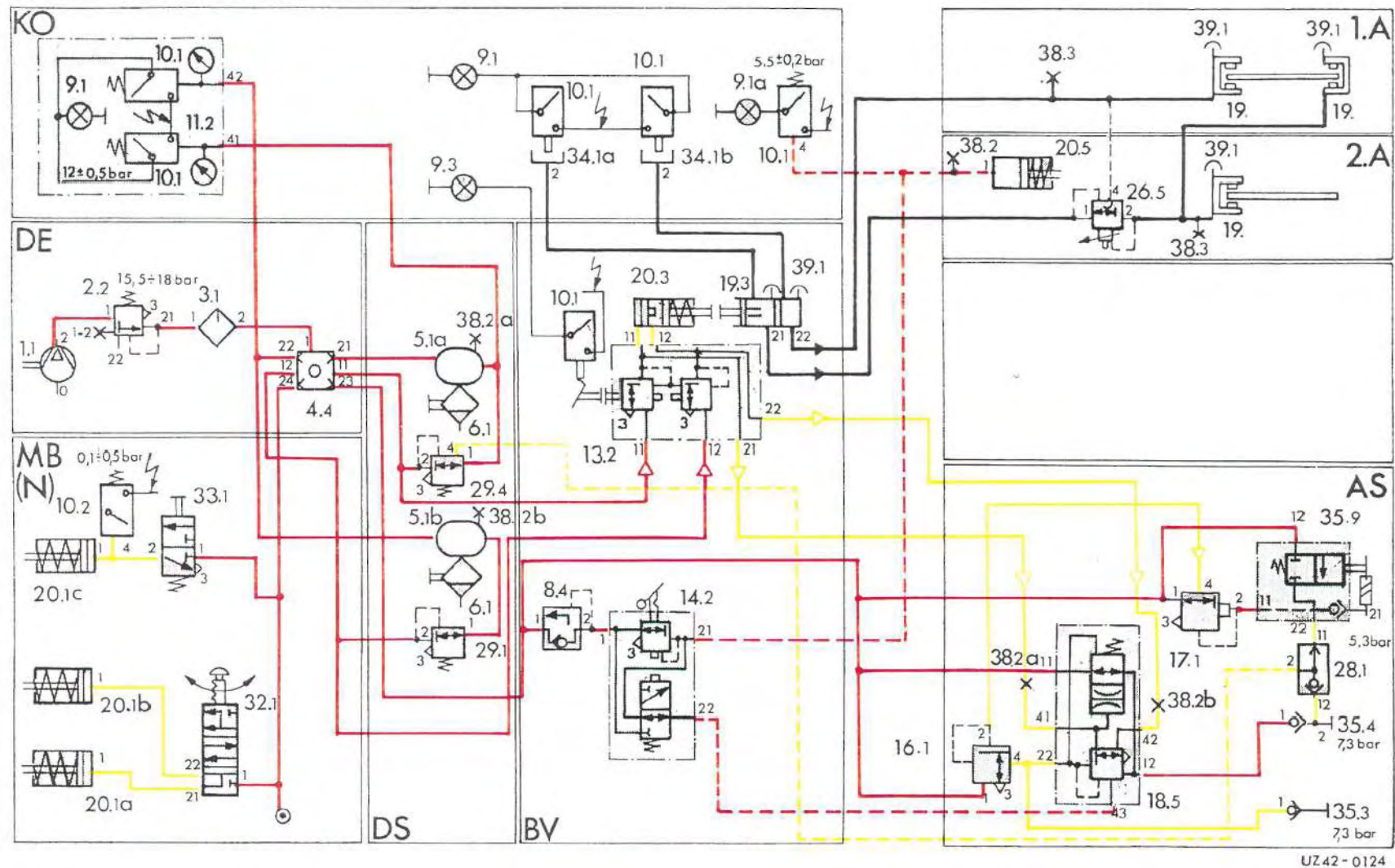
Dual-circuit hydraulic system and dual-circuit air brake system, ALB on rear axle spring-loaded parking brake and dual-line trailer brake system.

Refer to 43.11-1.5/1 to 1.5/28 for symbols.

Key to 1.5/3

A	Hydraulics	1.A	= 1st axle
B	Hydraulic control pressure	2.A	= 2nd axle
C	Brake pressure	AS	= Trailer control
D	Brake control pressure	BV	= Actuating valve
E	Supply pressure	DS	= Accumulator
F	Supply control pressure	MB(N)	= Engine brake (auxiliary load)
		DE	= Pressure generation
		KO	= Control switch
1.1	Air compressor	20.a	Piston cylinder 4-wheel drive
2.2	Pressure regulator with tyre inflating port	20.b	Piston cylinder differential lock
3.1	Antifreeze unit	20.1c	Piston cylinder engine brake
4.4	Four-circuit protection valve	20.3	Preload cylinder (dual-circuit)
5.1a	Air tank circuit 1	20.5	Spring-loaded parking brake
5.1b	Air tank circuit 2	26.5	Automatic brake pressure modulator (ALB)
6.1	Drain valve	29.1	Pressure reducing valve
8.1	Non-return valve	29.4	Pressure reducing valve with pneumatic control
9.1	Warning lamp, supply pressure	32.1	Changeover switch 4-wheel drive differential lock
9.1a	Indicator lamp, pressure difference (SA 35 975)	33.1	Changeover switch engine brake
9.1b	Indicator lamp parking brake	34.1	Equalizing reservoir
9.3	Stoplight	35.1b	Coupling head, front charging line
10.1	Electrical pressure switch supply pressure	35.3	Coupling head rear, brake
10.1a	Electrical pressure switch stoplight	35.4	Coupling head rear, supply
10.1b	Electrical pressure switch spring-loaded cylinder	38.2a	Test connection circuit 1
10.2	Electrical pressure switch engine brake	38.2b	Test connection circuit 2
10.3	Electrical pressure difference warning switch (SA 35 975)	38.2c	Test connection circuit 1
11.2	Double pressure gauge	38.2d	Test connection circuit 2
13.2	Service brake valve	38.2e	Test connection spring-loaded cylinder
14.2	Parking brake valve with brake test point	38.3	Test connection ALB modulator
18.5	Trailer control valve (2-line) with breakaway valve	39.1	Vent
19.a	Fixed combined caliper RA		
19.b	Fixed caliper FA circuit 1		
19.c	Fixed caliper FA circuit 2		
19.3	Master brake cylinder (dual-circuit)		

Functional Diagram (435.110/111) To chassis end No. 086 854



Dual-circuit hydraulic system and dual-circuit air brake system with ALB on rear axle and $\frac{1}{2}$ front axle, spring-loaded parking brake, single and dual-line trailer brake system.

Refer to 43.11-1.5/1 to 1.5/28 for symbols.

Key to 1.5/5

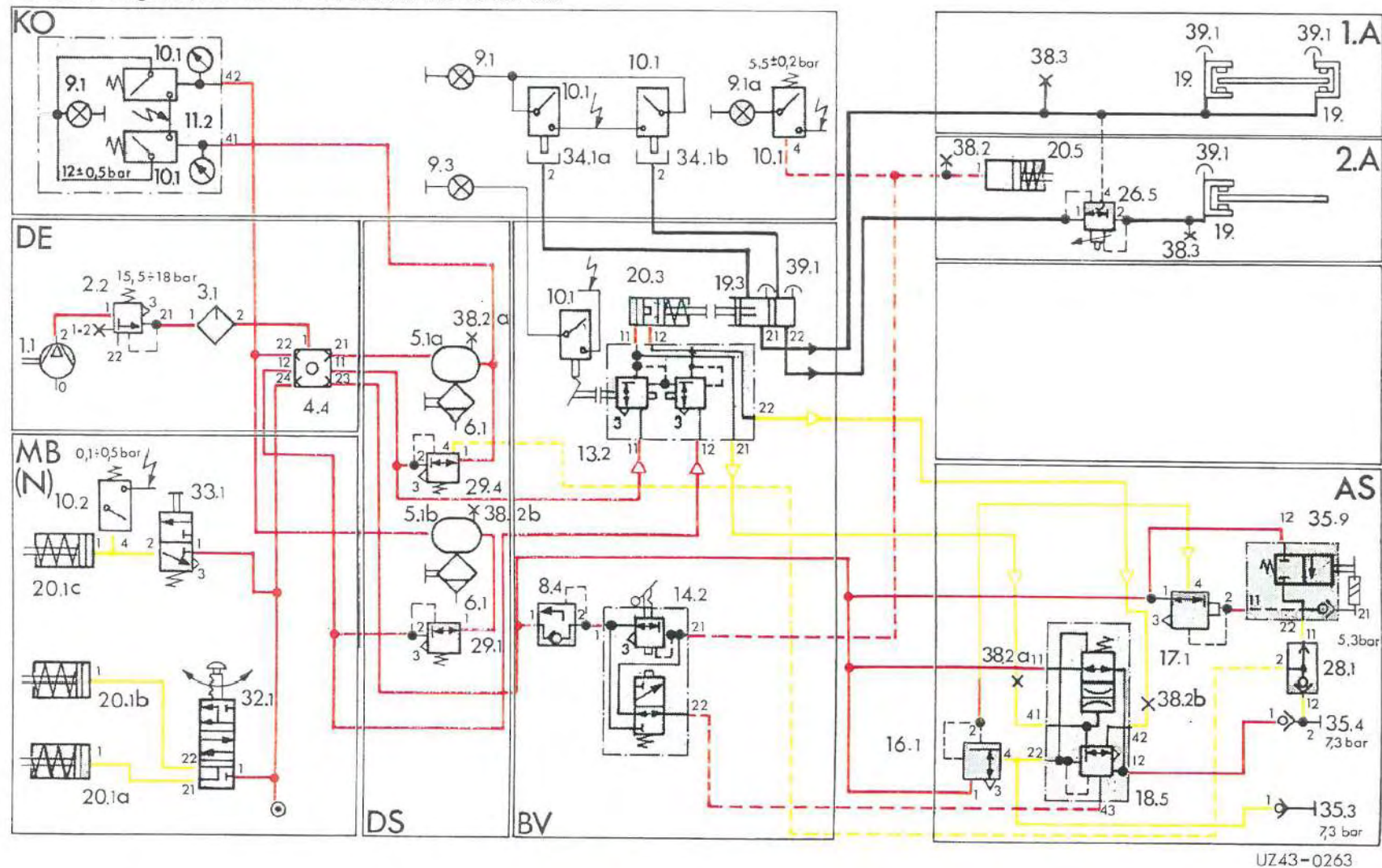
A Hydraulics
B Hydraulic control pressure
C Brake pressure
D Control pressure
E Supply
F Supply control pressure

1.A = 1st axle
2.A = 2nd axle
AS = Trailer control
BV = Actuating valve
DS = Accumulator
MB(N) = Engine brake (auxiliary load)
DE = Pressure generation
KO = Control switch

1.1 Air compressor
2.2 Pressure regulator
3.1 Antifreeze unit
4.4 Four-circuit protection valve
5.1a Air tank circuit 1
5.1b Air tank circuit 2
6.1 Water separator
8.4 Non-return valve with limited return flow
9.1 Warning lamp, red hydraulics
9.1a Warning lamp, red parking brake
9.3 Stoplight
10.1 Electrical pressure switch
10.2 Electrical pressure switch
11.2 Double pressure gauge
13.2 Service brake valve
14.2 Parking brake valve
16.1 Relay valve
17.1 Trailer control valve (1-line)
18.5 Trailer control valve (2-line)
19 Fixed caliper
19.3 Tandem master brake cylinder
20.1a Piston cylinder 4-wheel drive
20.1b Piston cylinder differential lock
20.1c Piston cylinder engine brake

20.3 Preload cylinder
20.5 Spring-loaded parking brake
26.5 ALB modulator
28.1 2-way valve trailer operation
29.1 Pressure reducing valve
29.4 Pressure reducing valve, switchover
32.1 Switchover valve, 4-wheel drive and differential lock
33.1 Switchover valve engine brake
34.1a Brake fluid reservoir circuit 1
34.1b Brake fluid reservoir circuit 2
35.3 Coupling head brake (2-circuit)
35.4 Coupling head supply (2-circuit)
35.9 Coupling head (single circuit)
38.2 Test connection spring-loaded cylinder
38.2a Test connection circuit 1
38.2b Test connection circuit 2
38.3 Test connection ALB modulator
39.1 Vent

Functional Diagram (435.110/111) From chassis end No. 086 855



Dual-circuit hydraulic system and dual-circuit air brake system, ALB on rear axle, spring-loaded parking brake, single and dual-circuit trailer brake system.

Refer to 43.11-1.5/1 to 1.5/28 for symbols.

UZ43-0263

Key to 1.5/7

A Hydraulics
B Hydraulic control pressure
C Brake pressure
D Control pressure
E Supply
F Supply control pressure

1.A = 1st axle
2.A = 2nd axle
AS = Trailer control
BV = Actuating valve
DS = Accumulator
MB(N) = Engine brake (auxiliary load)
D = Pressure generation
KO = Control switch

1.1 Air compressor
2.2 Pressure regulator
3.1 Antifreeze unit
4.4 Four-circuit protection valve
5.1a Air tank circuit 1
5.1b Air tank circuit 2
6.1 Water separator
8.4 Non-return valve with limited return flow
9.1 Warning lamp, red hydraulics
9.1a Warning lamp, red parking brake
9.3 Stoplight
10.1 Electrical pressure switch
10.2 Electrical pressure switch
13.2 Service brake valve
14.2 Parking brake valve
16.1 Relay valve
17.1 Trailer control valve (1-line)
18.5 Trailer control valve (2-line)
19 Fixed caliper
19.3 Tandem master brake cylinder
20.1a Piston cylinder 4-wheel drive
20.1b Piston cylinder differential lock
20.1c Piston cylinder engine brake

20.3 Preload cylinder
20.5 Spring-loaded parking brake
26.5 ALB modulator
28.1 2-way valve trailer operation
29.1 Pressure reducing valve
29.4 Pressure reducing valve, switchover
32.1 Switchover valve, 4-wheel drive and differential lock
33.1 Switchover valve, engine brake
34.1a Brake fluid reservoir, circuit 1
34.1b Brake fluid reservoir, circuit 2
35.3 Coupling head brake (2-circuit)
35.4 Coupling head supply (2-circuit)
35.9 Coupling head (single circuit)
38.2 Test connection spring-loaded cylinder
38.2a Test connection circuit 1
38.2b Test connection circuit 2
38.3 Test connection ALB modulator
39.1 Vent

The diagram is a detailed hydraulic schematic for a 4-way valve with 6 positions. It is divided into several functional blocks, each labeled with a code in the top left corner: KO, DE, MB, DS, BV, and AS. The schematic includes various hydraulic components such as pumps, valves, actuators, and flow control devices, each with its own set of ports and pressure ratings. The flow paths are indicated by solid and dashed lines, with red lines representing one path and yellow lines representing another. The diagram also shows the internal structure of the valve, including the spool and the various ports (1, 2, 3, 4, 11, 12, 21, 22, 23, 24, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 800, 801, 802, 803, 804, 805, 806, 807, 808, 809, 810, 811, 812, 813, 814, 815, 816, 817, 818, 819, 820, 821, 822, 823,

Refer to 43.11-1.5/1 to 1.5/28 for symbols